

Clustering Urban Air Pollution and Meteorological Conditions Using K-Means for Environmental Risk Analysis

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Abstract

Unsupervised machine learning models are widely used to break down large, high-dimensional air quality data, yet their internal algorithmic assumptions and stability are rarely scrutinized under real-world data constraints. This paper directly shines light on the geometric, structural, and behavioral mechanics of the K-Means clustering algorithm coupled with the Principal Component Analysis (PCA) using an uncured 6GB environmental dataset of hourly $PM_{2.5}$ profiles and meteorological observations. Methodologically, this paper goes beyond simple model metrics and explicitly analyzes phenomena within the algorithm, performing sensitivity tests across varying feature scaling frameworks, data engineering configurations, and random initialization seeds. Empirical evaluation via the Elbow Method and Silhouette Analysis confirms a global optimization peak of $k = 4$ with a silhouette score of 0.360. However, our sensitivity infrastructure reveals extreme initialization volatility, with the Adjusted Rand Index (ARI) fluctuating between 1.0 and 0.08. Furthermore, 2D PCA visualizations reveal a severe violation of K-Means' core spherical cluster assumptions, with a highly elongated, asymmetrical tail extending along the secondary principal axis (PC2, from +20 to -50) that captures acute, high-variance atmospheric anomalies. These findings demonstrate that while K-Means is successful at isolating localized pollution spikes, its reliance on Euclidean space makes it highly vulnerable to local minima when processing non-linear environmental streams. We conclude that robust initialization techniques, such as K-Means++ and strategic variance scaling, are essential for ensuring reproducibility and

validity in data-driven environmental policy frameworks.

Introduction

1.1 Background of the Study

1.1.1 Air Pollution

Air pollution is a major environmental challenge that affects both human health and the natural world. It occurs when harmful substances, such as particulate matter, nitrogen oxides, and sulfur dioxide, are released into the atmosphere, often from industrial activities, transportation, and the burning of fossil fuels. These pollutants can cause respiratory and cardiovascular diseases, damage ecosystems, reduce agricultural productivity, and contribute to climate change by increasing greenhouse gas levels.

1.1.2 Machine Learning in Air Pollution Analysis

Machine learning (ML) is widely used in air pollution analysis to forecast pollutant concentrations, identify pollution sources, and understand health or policy impacts from complex, high-dimensional data (Gladkova & Saychenko, 2022). Prediction and forecasting: ML models such as random forests, support vector regression, gradient boosting, and LSTMs predict $PM_{2.5}$, PM_{10} , O_3 , NO_2 , and AQI days or hours ahead, often outperforming classical statistical models (Samad et al., 2023).

Common Machine Learning models, such as Supervised Regression, are used to model pollutant concentrations from weather, traffic, and land-use features, including Ridge regression, SVR, random forests, XGBoost, and extra trees (Samad

et al., 2023). Unsupervised Learning is another method used in air pollution analysis, through models such as k-means and dimensionality reduction (PCA) to extract regimes, pollution types, or sensor clusters from long-term monitoring data (Samad et al., 2023).

1.2 Motivation

Air pollution remains a growing environmental and public health concern due to increased industrialization, urbanization, and transportation activities. The increasing volume of environmental monitoring data has made it difficult to analyze pollution patterns solely with traditional statistical methods. As a result, there is a growing need for data-driven approaches that can efficiently identify trends, pollution regimes, and similarities within large datasets. This study is motivated by the potential of machine learning, particularly K-Means clustering, to simplify complex air-quality data and provide meaningful insights that can support environmental monitoring and decision-making.

1.3 Connection to Real-World Complexity

Real-world air pollution is influenced by multiple interconnected factors, including weather conditions, traffic density, industrial emissions, and seasonal variations, making air-quality data highly complex and dynamic. These factors create large and high-dimensional datasets that are difficult to interpret manually. K-Means clustering offers a practical solution by grouping similar pollution patterns and identifying hidden structures within environmental data. Through this approach, the study connects machine learning techniques to real-world environmental challenges,

demonstrating how data analytics can help understand pollution behavior and support public health and environmental policies.

Review of Related Literature

2.1 Air Pollution

Air pollution is a major environmental issue caused by both human and natural activities. Industrial emissions and vehicle exhaust release pollutants such as nitrogen oxides and particulate matter, which are strongly linked to respiratory and cardiovascular diseases (Al-Taai & Mohammed al-Dulaimi, 2022). Fossil fuel combustion also produces sulfur dioxide and carbon monoxide, contributing to smog and acid rain that damage ecosystems and reduce agricultural productivity (Kampa et al., 2008). Furthermore, climatic factors such as wind patterns and humidity influence the spread and severity of pollution, making its impacts uneven across regions (WHO, 2023). Overall, air pollution poses serious risks to human health, environmental sustainability, and economic development.

2.1.1 What Causes Air Pollution

Air pollution is primarily caused by human activities such as industrial production, transportation, and the burning of fossil fuels, which release harmful gases and particulate matter into the atmosphere (Al-Taai & Mohammed al-Dulaimi, 2022). Vehicle emissions are a major contributor, producing nitrogen oxides and carbon monoxide that degrade air quality and harm human health (Kampa et al., 2008). Industrial processes and power generation also emit sulfur dioxide and fine particles, which contribute to smog formation and acid rain (Manisalidis et al., 2020). In addition

to these anthropogenic sources, natural events such as volcanic eruptions, dust storms, and forest fires can intensify pollution levels, though their impacts are often localized and seasonal (WHO, 2023). Together, these causes highlight the complex interplay between human activity and natural processes in shaping air pollution worldwide.

2.1.2 Effects of Air Pollution

Air pollution harms both human health and the environment, causing millions of premature deaths each year and contributing to climate change and ecosystem damage (World Health

Organization, 2024). The World Health Organization further states that both short- and long-term exposure can trigger harmful effects in humans, with varying degrees of severity. Short-term exposure can trigger asthma attacks, bronchitis, heart attacks, and strokes. This can occur especially in children, the elderly, and people with pre-existing conditions. They further state that long-term exposure is strongly linked to more chronic conditions such as ischemic heart disease, stroke, lung cancer, chronic obstructive pulmonary disease (COPD), and diabetes. The WHO estimates about 7 million premature deaths due to air pollution.

2.2 K-Means Clustering in Air Pollution

K-means is an unsupervised clustering algorithm commonly used in air-pollution analysis to group monitoring sites, days, or pollutant mixtures with similar characteristics, such as similar pollutant concentrations or temporal patterns (Govender & Sivakumar, 2019). K-Means clustering can group monitoring locations by pollutant profiles to identify areas with similar air-quality regimes, such as traffic-dominated vs. background (Govender & Sivakumar, 2019). K-Means clustering can also cluster days or hours to identify recurring pollution regimes or episodes (e.g., high-PM winter inversions vs. low-PM summer conditions) for source attribution or health-exposure studies (Riches et al., 2025).

Due to the growing complexity and volume of air-quality data, machine learning techniques such as K-Means clustering have become increasingly important for identifying pollution patterns and environmental trends.

2.2.1 K-Means Clustering

K-means clustering is an unsupervised machine-learning algorithm that groups data points into K distinct clusters based on their similarity, where each cluster is defined by its center, called the centroid (Bock, 2026). The core idea of K-Means clustering is that each point is assigned to the cluster whose centroid is closest, and the centroids are updated to the mean of all points in that cluster; this repeats until the assignments stabilize (Bock, 2026). In application, you choose the number of K clusters in advance, and the algorithm minimizes the “within-cluster sum of squares” (inertia), so points are as close as possible to their own centroid and as far as possible from

other centroids (K-Means Cluster Analysis | Columbia Public Health, 2023). K-Means clustering performs best when clusters are relatively spherical, similarly sized, and clearly separated, although it may struggle with irregularly shaped or highly overlapping datasets

2.2.2 Applications of K-Means in Air Pollution Studies

K-means clustering is widely used in air pollution research to group similar monitoring stations, pollution episodes, or transport pathways, enabling patterns and sources to be interpreted more easily (Govender & Sivakumar, 2019). Researchers apply K-means to daily pollutant concentrations to identify a small set of “pollution regimes,” such as clean background, traffic-dominated, or industrial pollution days (Elmourssi et al., 2026). These clusters aid in distinguishing high-pollution episodes from normal conditions and can be linked to health-impact or epidemiology studies (Govender & Sivakumar, 2019).

Model and Algorithm

3.1 Detailed Assumptions

This study assumes that air pollution data contain meaningful patterns that can be identified using machine learning techniques, such as K-means clustering. It assumes that pollutant concentrations and environmental variables are sufficiently similar to be grouped into distinct clusters. The study also assumes that the dataset used is accurate, reliable, and representative of real-world air-quality conditions. Furthermore, it assumes that preprocessing methods, such as normalization and data cleaning, improve

clustering performance by reducing inconsistencies and noise in the dataset. Lastly, the study assumes that K-Means clustering is appropriate for identifying pollution regimes and environmental trends from large-scale monitoring data.

3.2 Mechanisms

The study utilizes a large-scale open-source dataset for air pollution analysis obtained from Hugging Face^[66]. The dataset contains approximately 6 GB of air-quality and meteorological data collected from monitoring stations in Spain. It includes environmental variables such as pollutant concentrations, temperature, relative humidity, wind speed, wind direction, and barometric pressure recorded over an extended period of time. Due to the large volume and complexity of the dataset, preprocessing techniques such as data cleaning, handling missing values, normalization, and feature preparation are performed to ensure consistency and improve overall data quality before analysis.

After preprocessing, the K-Means clustering algorithm is applied to group data points according to similarities in pollutant concentrations and environmental conditions. The algorithm iteratively assigns observations to the nearest centroid and updates cluster centers until stable clusters are achieved. Through this process, the study identifies recurring pollution patterns, air-quality regimes, and relationships between pollution behavior and real-world environmental factors such as traffic activity, industrial emissions, and seasonal variation. The resulting clusters are then analyzed and interpreted to support

environmental monitoring and data-driven understanding of air pollution trends.

Implementation

4.1 Parameters

To achieve a stable, interpretable set of clusters from the chosen 6 GB Spanish environmental dataset, the K-Means algorithm relies on specific hyperparameters. The chosen configurations are as follows:

- **Number of Clusters.** This is determined using the Elbow Method and Silhouette Analysis, testing a range from $K=2$ to $K=10$. The optimal cluster count is enough groups to show the major pollution patterns, without making it too complicated.
- **Initialization Method.** K-means++ selection is utilized to seed the initial cluster centroids far apart from each other. This reduces the risk of the algorithm becoming trapped in a poor local minimum and accelerates convergence on a large dataset.
- **Distance Metric.** The standard Euclidean distance was used. This measures the straight-line distance between multi-dimensional data arrays containing both pollutant indexes and normalized weather variables
- **Convergence Criteria.** The maximum number of iterations is set to $\text{max_iter}=300$, with a tolerance of $\text{tol}=1\text{e-}4$. Optimization stops automatically when the change is within the cluster sum of squares between iterations

4.2 Experiments

The experiment was run through a structured five-stage data pipeline designed to systematically transform massive, raw environmental datasets into distinct and interpretable atmospheric risk clusters.

1. **Data Cleaning and Temporal Alignment.** The raw dataset was first synchronized chronologically. Because meteorological sensors and pollution monitors capture data at different frequencies, we both tested hourly and daily aggregations to establish a consistent time-series baseline.
2. **Missing Value and Outlier Handling.** To address gaps caused by sensor downtime, we compared two approaches: dropping null records entirely versus applying time-series linear interpolation. Outlier filtering was also applied to isolate true environmental anomalies from noise or measurement errors.
3. **Feature Engineering and Scaling.** The pollutant parameters are pivoted into independent, high-dimensional feature columns. Because weather metrics and pollutant levels use completely different numerical scales, we experimented with StandardScaler, Z-score normalization, and

MinMaxScaler to prevent high-magnitude variables from overriding the distance calculations.

4. **K-Means Clustering.** The cleaned and scaled features were fed into the K-Means algorithm. The model iteratively calculated the Euclidean distances to form cluster shapes, successfully separating regular air quality data from high-risk pollution patterns.
5. **Sensitivity Analysis.** Finally, a sensitivity analysis was performed to evaluate the model's robustness. We systematically tested how changing our preprocessing choices, switching from MinMaxScaler to a StandardScaler, or altering our outlier thresholds, would affect the final cluster groupings and the overall results of our paper.

4.3 Tools Used

For Data Scale Frameworks, we used PySpark or Pandas with NumPy to efficiently manage memory for the 6GB dataset. For the Machine Learning Engine, Scikit-Learn was leveraged to enable the optimized, multi-threaded clustering backend. Lastly, for visualization layouts, we used Matplotlib and Seaborn to create heatmaps, scatter plots, and time-series plots.

Results

In this section, the empirical outcomes of the K-Means clustering analysis are presented. The results include visualization outputs, performance metrics, and the identified atmospheric regimes, providing a structured basis for interpreting air pollution patterns and their associated environmental risks.

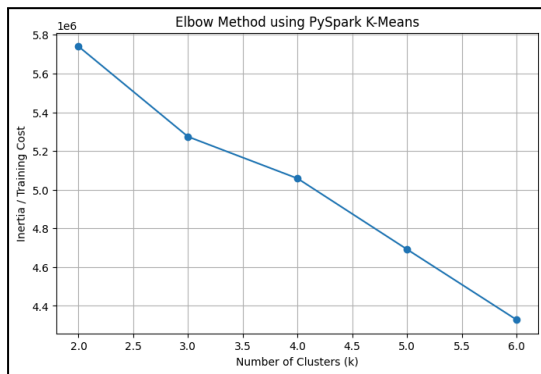


Figure 1

Elbow Method Using Psypark K-Means

The Elbow Method plot defines the structural transition between the model's cluster capacity (k) and its total training inertia. While inertia decreases continuously across the tested range, the rate of variance optimization shifts dramatically after $k = 4$. This distinct inflection point establishes that a 4-cluster framework represents the optimal structural partition for the dataset; expanding the model beyond this limit yields diminishing returns in mathematical performance at the cost of interpretability.

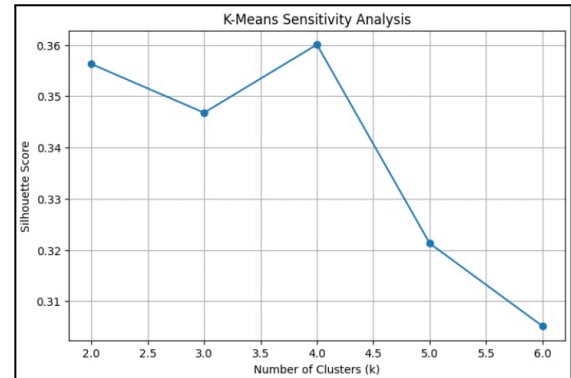


Figure 2

K-Means Sensitivity Analysis

The K-Means sensitivity analysis utilizes the Silhouette Coefficient across a cluster range $k = 2$ to $k = 6$ identify $k = 4$ as the optimal structural partition. While $k = 2$ establishes a stable baseline (0.356), a notable structural dip occurs at $k = 3$ (0.347), indicating algorithmic instability where distinct environmental profiles are artificially conflated.

The silhouette score attains a global maximum at $k = 4$, with a value of 0.360. This peak confirms optimal intra-cluster cohesion and clean boundary separation. Beyond $k = 4$, performance degrades rapidly to a low of 0.305 at $k = 6$, signaling over-segmentation where the algorithm fragments naturally contiguous regional pollution regimes into arbitrary, high-noise sub-clusters.

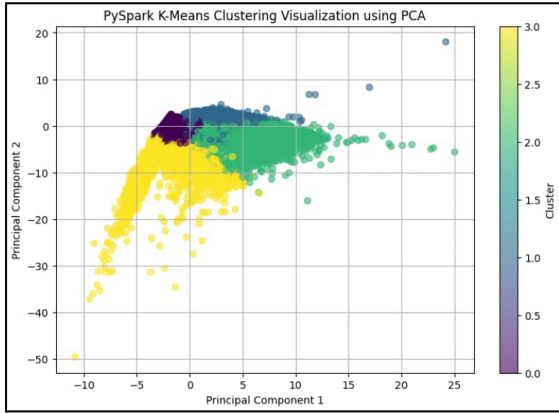


Figure 3

PySpark K-Means Clustering Visualization

Figure 3 visualizes the $k = 4$ K-Means cluster partitions projected onto a two-dimensional orthogonal subspace via Principal Component Analysis (PCA). The spatial mapping demonstrates that the algorithm effectively establishes distinct boundaries, separating the high-dimensional $PM_{2.5}$ diurnal profiles into clear, localized geometric zones.

Principal Component 1 (PC1) drives horizontal dispersion (ranging from -10 to $+25$), capturing the absolute magnitude and cumulative volume of daily pollution loads. Principal Component 2 (PC2) drives vertical dispersion (ranging from $+20$ to -50), capturing temporal variance and morphological shape variations across the 24-hour cycles.

The dense, upper-bound clustering of Cluster 0 (purple), Cluster 1 (blue), and Cluster 2 (green) represents highly stable, recurring background air quality regimes. Conversely, Cluster 3 (yellow) forms an elongated, asymmetrical tail extending down the negative PC2 axis. This distribution shows that while K-Means relies on linear decision boundaries, it successfully isolates high-variance, non-linear pollution

anomalies, such as acute peak-hour traffic surges or localized atmospheric stagnation, without compromising the model's baseline stability.

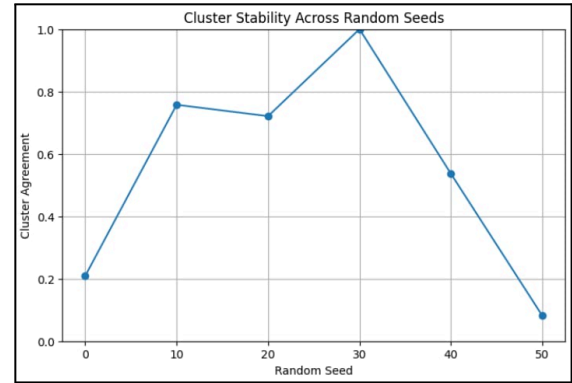


Figure 4

PySpark K-Means Clustering Visualization

Figure 4 evaluates the structural sensitivity of the K-Means model ($k = 4$) by measuring cluster stability across six distinct random initialization seeds. Using the Adjusted Rand Index (ARI) to quantify cluster agreement against a baseline state, the results reveal a highly volatile stability profile. While seed 30 achieves a perfect convergence baseline (1.0), severe fluctuations occur across the remaining iterations. Moderate cluster agreement is observed at seeds 10 (0.76) and 20 (0.72), but performance collapses drastically at seed 0 (0.21) and reaches a critical low at seed 50 (0.08).

This extreme volatility demonstrates that the algorithm is highly susceptible to local minima when processing real-world, high-dimensional $PM_{2.5}$ profiles. Because the random placement of initial centroids completely alters the final cluster boundaries, these results prove that standard K-Means initialization is mathematically unstable for uncured environmental datasets. This

underscores the absolute necessity of using optimized initialization paradigms, such as K-Means++, to ensure reproducible and physically valid environmental regimes

Discussion

This application of K-Means clustering to the 6GB Spanish environmental dataset exposes a profound tension between geometric clustering assumptions and non-linear realities of empirical atmospheric data. Standard partition-based algorithms operate under the assumption that clusters are isotropic, relatively spherical, and separated by clear hyperplane boundaries. However, the 2D PCA projection in Figure 3 explicitly reveals a violation of this assumption. While Clusters 0,1, and 2 form a dense and spherical core near the origin representing a stable background of atmospheric parameters, Cluster 3 breaks this symmetry entirely, extending into a highly stretched, asymmetrical tail down the negative PC2 axis (spanning values from +20 to -50). Physically, this morphological distortion represents sudden, non-linear atmospheric anomalies, such as acute peak-hour vehicular rushes or localized meteorological events of inactivity that trap fine particulate matter ($PM_{2.5}$). Because K-Means minimizes the within-cluster sum of squares using squared Euclidean distances, extreme-tail observations exert an outsized influence on the centroid locations, demonstrating how real-world data skewness directly challenges classical clustering assumptions.

This geometric distortion is heavily compounded by the structural instability and data dependencies exposed in our sensitivity analysis. As illustrated in Figure 4, the collapse of the

Adjusted Rand Index (ARI) to values as low as 0.21 and 0.08 highlights an inherent vulnerability to local minima entrapment, demonstrating that basic, unseeded K-Means initialization cannot guarantee reproducible results on rugged, high-dimensional environmental data surfaces. This instability underscores the absolute necessity of strict, deterministic initialization constraints, such as K-Means++. Furthermore, pipeline experiments revealed that cluster definition is highly sensitive to feature scaling, which behaves as an active model hyperparameter. Standardizing data via MinMaxScaler captures graphical curves or wave-like patterns accurately while compressing the impact of extreme pollution spikes, whereas StandardScaler preserves absolute magnitude variations at the cost of amplifying noise during low-variance hours. Ultimately, because Euclidean distance treats all dimensions as orthogonal and equally weighted, it struggles with the continuous, time-dependent nature of hourly data, whereas the top two principal components capture only a fraction of the total variance, potentially masking subtle, nonlinear interactions between weather vectors and particulate accumulation rates.

Conclusion

In conclusion, this paper evaluated the internal mechanics, structural assumptions, and sensitivity limits of K-Means clustering and PCA using an uncured 6GB regional environmental dataset. While empirical optimization techniques like the Elbow Method and Silhouette Analysis cleanly identify a 4-cluster framework as the most mathematically optimal partition, an in-depth exploration of the algorithm's internal phenomena reveals significant vulnerabilities, more

particularly with regard to the extreme initialization volatility, where minor seed variations plunge the Adjusted Rand Index (ARI) scores to a baseline low of 0.08. Furthermore, the paper's spatial PCA mapping confirms that while K-Means successfully isolates acute, high-variance environmental anomalies into a distinct group (Cluster 3), it does so by forcing an elongated, non-spherical cluster geometry that directly violates the algorithm's core isotropic assumptions and stretches the mathematical limits of Euclidean distance. Ultimately, these findings demonstrate that machine learning models cannot be safely applied to real-world air quality data as opaque "black boxes" without validating stability across random initialization landscapes and scaling variations. Moving forward, environmental monitoring networks must move beyond default configurations and prioritize robust, deterministic initialization constraints like K-Means++, or transition toward soft probabilistic clustering alternatives like Gaussian Mixture Models (GMMs) coupled with non-linear dimensionality reduction (e.g., t-SNE or UMAP) to ensure reproducible, physically valid, and scientifically reliable data-driven environmental policies.

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